Mateo del Rio Lanse
US Citizen | md2292@cornell.edu | (914) 758-2446
Portfolio: mdelriolanse.com | LinkedIn: mdelriolanse | GitHub: mdelriolanse

EDUCATION

Cornell University, College of Engineering **Ithaca, NY**
B.S. Electrical & Computer Engineering (GPA: 3.75 / 4.00) *Expected Graduation: Dec. 2028*
Relevant Coursework: CS 4414 (Systems Programming), ECE 3140 (Embedded Systems), ECE 2300 (Digital Logic and Computer Organization), MATH 2940, PHYS 2213, MATH 2930

PROFESSIONAL EXPERIENCE

Options Technology **New York, NY**
Software Engineering Intern, PrivateMind Platform *Jun. 2026 – Aug. 2026*

- Owned and shipped an OpenAI-compatible Batch API end-to-end (async worker draining against GPU capacity, per-request billing, DB schema, SDK, Helm deployment), backed by ~7,000 lines of contract tests; integrated Pure Storage FlashBlade S3 results storage for large inference artifacts.
- Benchmarked frontier LLMs before production serving: Poisson-distributed concurrency sweeps measuring time-to-first-token, inter-token latency, end-to-end latency, pre-emption, and throughput at p50/p95/p99 across multi-GPU HGX deployments.
- Owned the gateway load-management layer (priority scheduling, per-model streaming timeouts/keepalive, per-key concurrency limits, token-budget admission control gating batch work on GPU memory pressure) on NVIDIA B200/B300 GPUs.
- Brought a novel text-to-image model through full production deployment (GPU configuration, gateway routing, bounded concurrency, per-image pricing, UI) and built the platform content-safety layer.
- Merged 133 PRs across 10 repos (Rust, TypeScript, Python, SQL, Helm); unified two divergent agent execution paths, deleting ~6,000 lines of legacy code.

Cornell University **Ithaca, NY**
AI Alignment Undergraduate Researcher, Cornell NLP Group *Oct. 2025 – May 2026*

- Built the evaluation pipeline behind "Open Weight, Open Risk" (co-author): 15 frontier open-weight models across 800 WMDP-derived adversarial bio/chem requests, run locally on B200-class GPUs with vLLM and SGLang.
- Co-designed the training-free "inception" attack, in which a small uncensored 7B architect model pivots target reasoning and lifts compliance from a ~10% baseline to >95% under 5 iterations, exposing "shallow alignment" in open-weight reasoning models, with a PhD researcher at the Cornell NLP Group.

Engineering Manager & ML Engineer, Generative AI at Cornell *Oct. 2025 – May 2026*

- Led an 8-person engineering team building an ESG risk monitoring platform for Investcorp ($60B+ AUM).
- Implemented a vendor due-diligence agentic pipeline with fintech company QuickFi.
- Collected vendor data from government websites, regulatory filings, and public databases; reconciled it into consolidated profiles and generated written due-diligence reports.

Hospital for Special Surgery **New York, NY**
Machine Learning Research Intern *Jun. 2024 – Aug. 2024*

- Analyzed immunological datasets with Dr. Amit Lakhanpal and generated protein embeddings using Meta's ESM model to support rheumatoid arthritis treatment research.

PROJECTS

Dispatch *Pitch Deck* | *GitHub*

- Built an agentic pentesting platform that converts vulnerability findings into ready-to-merge GitHub PRs.
- Orchestrated security agents using Mastra and OpenRouter with isolated execution via Blaxel Sandboxes, and Slackbot/Datadog middleware for scan triggering and log ingestion.

Debately *Website* | *GitHub*

- Built a moderated debate platform with fact-checking pipelines and an LLM-driven consensus tracker (Anthropic API, Supabase, Vercel/Railway).
- Engineered fact-checking pipelines with web data normalization, preventing hallucinations in consensus.

TensorAtelier *GitHub*

- Built a pluggable PyTorch interface with automatic optimization/profiling and configurable multi-accelerator support, reducing boilerplate in ML research.

AWARDS

Cornell AI Hackathon (2026) — Dispatch, Best Developer Tool
2x Regeneron WESEF (2024 & 2025) — 2nd Place in Computer Science

SKILLS

Languages: TypeScript, Rust, C, Python
Technologies: Linux, Kubernetes, Openshift, Helm, vLLM, SGLang, Pure Storage FlashBlade S3, Supabase, PostgreSQL, Vercel, React, Cloudflare